A singular endpoint in the Rényi–Sobolev asymptotic-tightness theorem

Counterexample/correction for arXiv:2306.12288

13 August 2026

Proof intuition

Order-zero entropy measures only how much of the space is in the support. To make it positive, a function must have a hard zero. For $q > 1$, the factor f^{q-1} also vanishes at that zero, so the edge energy stays finite. Exactly at $q = 1$, the factor turns into $\ln f$ and a hard zero becomes an infinite barrier. Below one it becomes a negative power and is singular again. The endpoint is therefore a genuine phase transition hidden by the notation "$q \geq 1$".

Endpoint counterexample. *For the binary Bonami–Beckner semigroup and every $n \geq 1$,*

$$\Xi_{0,1}^{(n)}(\alpha) = +\infty \qquad (0 < \alpha < \ln 2).$$

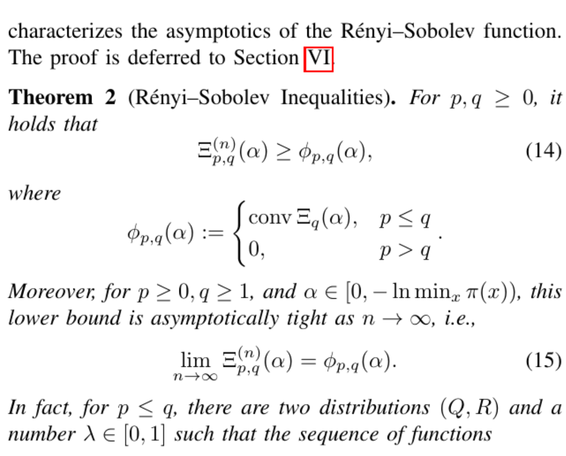
Thus Theorem 2 of the source is false at the explicitly included endpoint $(p, q) = (0, 1)$: it claims convergence to the finite number

$$\Xi_1(\alpha) = \left(\frac{1}{2} - y\right) \ln \frac{1-y}{y}, \qquad y = h^{-1}(\ln 2 - \alpha) \in (0, \tfrac{1}{2}).$$

The theorem's lower bound remains true; its claimed asymptotic tightness at this endpoint does not.

The exact claim in the source

Theorem 2 says "for $p \geq 0, q \geq 1$" and therefore includes $p = 0, q = 1$. Its binary specialization again asserts asymptotic tightness throughout $p \geq 0, q \geq 1$ and $0 \leq \alpha < \ln 2$. The source's own formula (17) gives the finite value of Ξ_1 displayed above.



characterizes the asymptotics of the Rényi–Sobolev function. The proof is deferred to Section VI.

Theorem 2 (Rényi–Sobolev Inequalities)**.** *For $p, q \geq 0$, it holds that*

$$\Xi_{p,q}^{(n)}(\alpha) \geq \phi_{p,q}(\alpha), \tag{14}$$

where

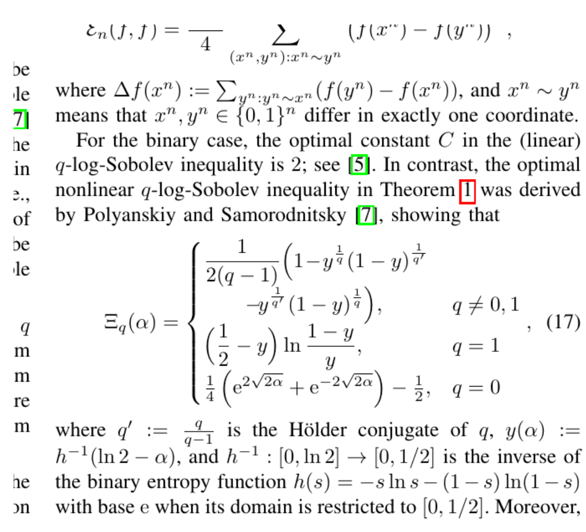
$$\phi_{p,q}(\alpha) := \begin{cases} \operatorname{conv} \Xi_q(\alpha), & p \leq q \\ 0, & p > q \end{cases}.$$

Moreover, for $p \geq 0, q \geq 1$, and $\alpha \in [0, -\ln \min_x \pi(x))$, this lower bound is asymptotically tight as $n \to \infty$, i.e.,

$$\lim_{n\to\infty} \Xi_{p,q}^{(n)}(\alpha) = \phi_{p,q}(\alpha). \tag{15}$$

In fact, for $p \leq q$, there are two distributions (Q, R) and a number $\lambda \in [0, 1]$ such that the sequence of functions

Theorem 2, official arXiv PDF page 4.



where $\Delta f(x^n) := \sum_{y^n : y^n \sim x^n} (f(y^n) - f(x^n))$, and $x^n \sim y^n$ means that $x^n, y^n \in \{0,1\}^n$ differ in exactly one coordinate.

For the binary case, the optimal constant C in the (linear) q-log-Sobolev inequality is 2; see [5]. In contrast, the optimal nonlinear q-log-Sobolev inequality in Theorem 1 was derived by Polyanskiy and Samorodnitsky [7], showing that

$$\Xi_q(\alpha) = \begin{cases} \dfrac{1}{2(q-1)}\Big(1 - y^{\frac{1}{q}}(1-y)^{\frac{1}{q'}} \\ \qquad - y^{\frac{1}{q'}}(1-y)^{\frac{1}{q}}\Big), & q \neq 0, 1 \\ \left(\dfrac{1}{2} - y\right) \ln \dfrac{1-y}{y}, & q = 1 \\ \frac{1}{4}\left(\mathrm{e}^{2\sqrt{2\alpha}} + \mathrm{e}^{-2\sqrt{2\alpha}}\right) - \frac{1}{2}, & q = 0 \end{cases}, \tag{17}$$

where $q' := \frac{q}{q-1}$ is the Hölder conjugate of q, $y(\alpha) := h^{-1}(\ln 2 - \alpha)$, and $h^{-1} : [0, \ln 2] \to [0, 1/2]$ is the inverse of the binary entropy function $h(s) = -s \ln s - (1-s)\ln(1-s)$ with base e when its domain is restricted to $[0, 1/2]$. Moreover,

Finite binary formula (17), official PDF page 4.

Proof of the counterexample

Write $Q_n = \{0,1\}^n$ and let π^n be uniform measure. The source defines the order-zero two-parameter entropy of every nonnegative function by

$$\mathrm{Ent}_{0,q}(f) = -\ln \pi^n(f > 0).$$

Let f be nonzero and satisfy $n^{-1}\,\mathrm{Ent}_{0,1}(f) \geq \alpha > 0$. If $S = \{x : f(x) > 0\}$, then

$$\ln \frac{2^n}{|S|} \geq n\alpha > 0.$$

Hence S is nonempty and proper. Since the hypercube is connected, an edge $x \sim y$ crosses its boundary. Relabel it so that $f(x) = a > 0$ and $f(y) = 0$.

At $q = 1$ the source uses the continuous endpoint

$$\frac{1}{n}\,\frac{\mathcal{E}_n(f, \ln f)}{\mathbb{E}_{\pi^n} f}.$$

To evaluate it at a function with zeros, replace every zero of f by $\varepsilon > 0$. The symmetrized edge formula for the Dirichlet form contains, on the selected boundary edge, a fixed positive multiple of

$$(a - \varepsilon)(\ln a - \ln \varepsilon).$$

All edge summands are nonnegative because the logarithm is increasing. The displayed term tends to $+\infty$ as $\varepsilon \downarrow 0$, whereas $\mathbb{E} f_\varepsilon \to \mathbb{E} f > 0$. Therefore the endpoint energy of every feasible f is $+\infty$.

This proves $\Xi^{(n)}_{0,1}(\alpha) = +\infty$. The constraint is not vacuous: a singleton support has entropy $n \ln 2$, so it is feasible for every $\alpha < \ln 2$. If instead one interprets the $q = 1$ endpoint as admitting only strictly positive f, the conclusion is identical by the standard extended-real convention: every such f has $\mathrm{Ent}_{0,1}(f) = 0$, so the positive-$\alpha$ feasible class is empty and its infimum is $+\infty$.

For comparison, when $0 < \alpha < \ln 2$ the source formula has $0 < y < 1/2$, hence $(1/2 - y)\ln((1-y)/y)$ is a finite positive real number. An identically infinite sequence cannot converge to it.

One-bit calculation

The singularity is already visible before taking products. For $f_\varepsilon = (1, \varepsilon)$, the source's one-bit generator gives

$$\frac{\mathcal{E}(f_\varepsilon, \ln f_\varepsilon)}{\mathbb{E} f_\varepsilon} = \frac{1-\varepsilon}{2(1+\varepsilon)} \ln \frac{1}{\varepsilon} \longrightarrow +\infty.$$

This is the complete obstruction in its smallest form.

Where the proof fails

The asymptotic-tightness proof reduces to $p = 0$ and conditions a product distribution on a typical set. That distribution has zeros. Fact 1 then asserts a uniform energy bound for every distribution and every $q \geq 1$, on the ground that the expression is continuous on the closed probability simplex. At $q = 1$ this is false.

We next need the following fact.

Fact 1. *For* $q \geq 1$, $\mathcal{E}\left(\left(\frac{S}{\pi}\right)^{1/q}, \left(\frac{S}{\pi}\right)^{1/q'}\right)$ *is uniformly bounded for any distribution* S.

6) This fact is obvious since it holds that

$$\mathcal{E}\left(\left(\frac{S}{\pi}\right)^{1/q}, \left(\frac{S}{\pi}\right)^{1/q'}\right)$$
$$= -\sum_{x,y} L_{x,y}\left(\frac{S(y)}{\pi(y)}\right)^{1/q}\left(\frac{S(x)}{\pi(x)}\right)^{1/q'}\pi(x).$$

The RHS above is continuous in S on the probability simplex.

7) So, the term $S \mapsto \mathcal{E}\left(\frac{S}{\pi}, \ln\frac{S}{\pi}\right)$ is bounded.

By the fact above, there is a number M such that for any

m S,

$$\left|\mathcal{E}\left(\left(\frac{S}{\pi}\right)^{1/q}, \left(\frac{S}{\pi}\right)^{1/q'}\right)\right| \leq M. \tag{51}$$

Fact 1 and its continuity claim, official arXiv PDF page 11.

Indeed, on the binary space take $S_\varepsilon = (1-\varepsilon, \varepsilon)$. Since $\pi = (1/2, 1/2)$, direct substitution gives

$$\mathcal{E}\left(\frac{S_\varepsilon}{\pi}, \ln\frac{S_\varepsilon}{\pi}\right) = \frac{1}{2}(1-2\varepsilon)\ln\frac{1-\varepsilon}{\varepsilon} \longrightarrow +\infty.$$

Consequently the constant M invoked immediately after Fact 1 does not exist at $q = 1$.

Nor can the construction be repaired by smoothing the typical-set law. Adding any positive background multiple of π^n makes the law have full support, and its order-zero Rényi divergence becomes exactly zero; it then fails every constraint with $\alpha > 0$. Retaining any zero leaves the boundary singularity. This support-versus-energy dichotomy is structural.

The proposed $q < 1$ extension

The source leaves $q < 1$ for future work. The same calculation shows that such an extension cannot retain the endpoint $p = 0$. For $0 < q < 1$, a boundary edge contributes after regularization a positive multiple of

$$\frac{(a-\varepsilon)(a^{q-1} - \varepsilon^{q-1})}{q-1} \longrightarrow +\infty.$$

Thus $\Xi_{0,q}^{(n)}(\alpha) = +\infty$ under the natural extended definition for $\alpha > 0$. This observation does not address the genuinely different range $p > 0$.

Scope and review

Scope. This is a full counterexample to the asymptotic-tightness assertion at the stated endpoint $(p, q) = (0, 1)$, already for the source's binary example and for every finite n. It does not challenge the lower bound, the proven range $q > 1$, or the future-work range $p > 0, q < 1$.

Novelty check. Cheap run indexes and bounded exact-title, theorem-phrase, endpoint, erratum, and correction searches through 13 August 2026 found no duplicate or published correction. The official arXiv v2 PDF dated 29 July 2024 was used here. Independent human review remains appropriate.

Human review. Check only three points: the explicit inclusion of $p = 0, q = 1$ in Theorem 2, the continuous-extension convention for $\mathcal{E}(f, \ln f)$ at a support boundary, and the factor in the one-bit generator. None affects the qualitative divergence.

References

[1] L. Yu, *Rényi–Sobolev Inequalities and Connections to Spectral Graph Theory*, arXiv:2306.12288v2 (2024), Theorem 2 and Fact 1.